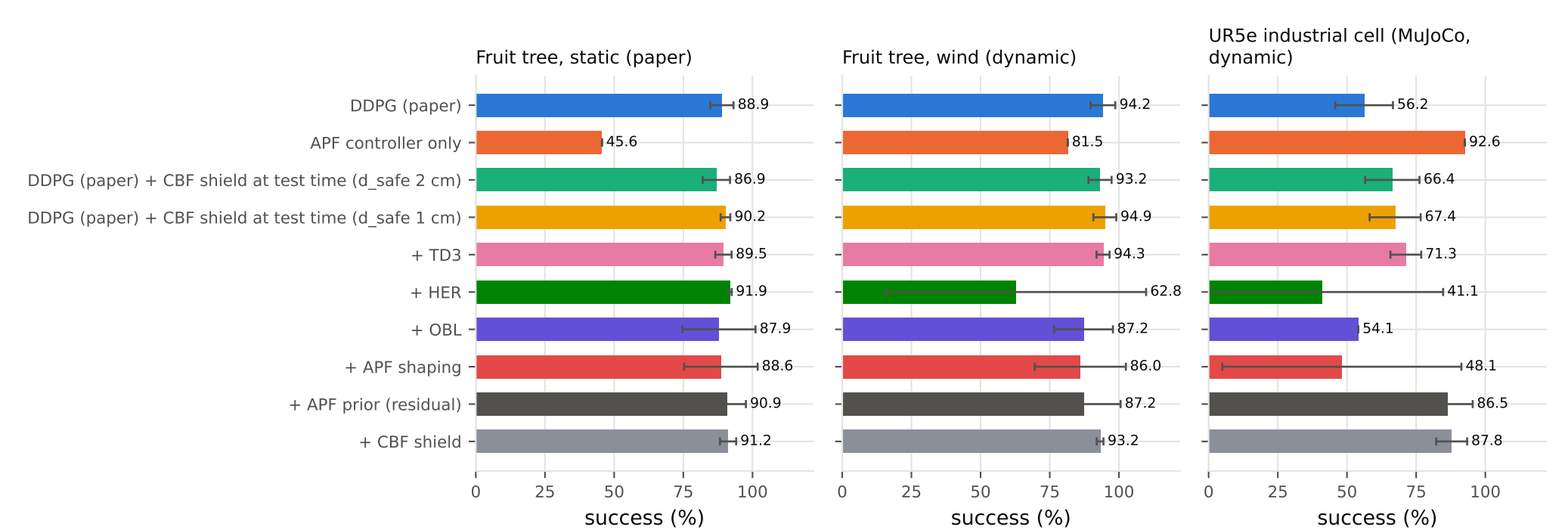
Fruit tree, static (paper)
Fruit tree, wind (dynamic)
UR5e industrial cell (MuJoCo, dynamic)
DDPG (paper)
APF controller only
DDPG (paper) + CBF shield at test time (d_safe 2 cm)
DDPG (paper) + CBF shield at test time (d_safe 1 cm)
+ TD3
+ HER
+ OBL
+ APF shaping
+ APF prior (residual)
+ CBF shield
88.9
45.6
86.9
90.2
89.5
91.9
87.9
88.6
90.9
91.2
94.2
81.5
93.2
94.9
94.3
62.8
87.2
86.0
87.2
93.2
56.2
92.6
66.4
67.4
71.3
41.1
54.1
48.1
86.5
87.8
0
25
50
75
100
success (%)